

ASSIGNMENT -04

Take one domain and build business Understanding

Domain: Healthcare – Disease Prediction Using Machine Learning

1. Introduction to the Domain

Healthcare is one of the most data-intensive domains in the world. Hospitals, diagnostic centers, and clinics generate massive volumes of data every day in the form of patient records, lab reports, prescriptions, imaging results, and monitoring data. With the growth of Electronic Health Records (EHRs), wearable devices, and IoT-based health monitoring systems, the healthcare sector has entered the era of Big Data.

In this report, we focus on **Disease Prediction using Machine Learning** as a business domain. The objective is to understand the business problem, stakeholders, goals, constraints, risks, and success criteria before developing a technical solution.

Business understanding is the first and most critical phase of the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology. Without clear business clarity, even technically advanced models fail to deliver value.

2. Problem Statement

Healthcare institutions face the following challenges:

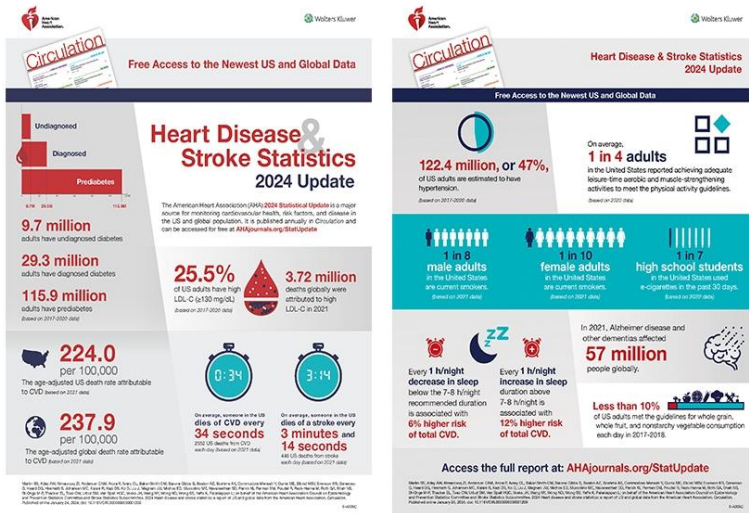
- Late diagnosis of chronic diseases
- High treatment costs due to delayed detection
- Overburdened doctors
- Increasing patient population
- Lack of early risk identification

Example Problem:

Predict whether a patient is at risk of developing **heart disease** based on historical medical data.

The goal is not just technical prediction but **improving early diagnosis and reducing mortality rates**.

3. Industry Context



Cardiovascular diseases are among the leading causes of death globally. Hospitals spend significant resources managing emergency cases that could have been prevented with early intervention.

Business context includes:

- Rising healthcare costs
- Demand for preventive healthcare
- Government regulations on patient data
- Insurance company risk assessment
- Competition among hospitals for quality services

Hospitals are now investing in AI-based decision-support systems to improve patient care and operational efficiency.

4. Business Objectives

The main business objectives are:

1. **Early Disease Detection**
 - Identify high-risk patients before symptoms become severe.
2. **Reduce Treatment Costs**
 - Prevent expensive emergency procedures.

3. Improve Patient Outcomes

- Increase survival rates and quality of life.

4. Enhance Hospital Reputation

- Better diagnosis accuracy improves trust.

5. Optimize Resource Allocation

- Reduce unnecessary lab tests and hospital admissions.

These objectives must align with organizational strategy and long-term growth plans.

5. Stakeholder Analysis

Different stakeholders are involved in this domain:

Stakeholder	Role	Expectations
Doctors	Use prediction system	Accurate and explainable results
Patients	Receive care	Early detection and affordable treatment
Hospital Management	Decision makers	Cost reduction & ROI
IT Department	System integration	Secure & scalable solution
Insurance Companies	Risk evaluation	Reliable risk prediction
Government	Regulatory body	Compliance with health laws

Understanding stakeholder needs ensures the solution delivers real business value.

6. Business Requirements

Functional Requirements

- Predict probability of disease.
- Provide risk score (Low/Medium/High).
- Generate patient risk report.
- Integration with hospital database.

Non-Functional Requirements

- Data security and privacy (HIPAA-like compliance)
- High accuracy (>85%)
- Real-time response
- Scalable architecture

Legal & Ethical Requirements

- Patient data confidentiality
- No bias based on gender, age, or ethnicity
- Transparent decision-making

7. Data Understanding from Business Perspective

studies

Data source	Sample size	References
Single psychiatric inpatient unit	728–2,010	82, 97
Specialized center/clinic	544–10,017	15, 40
Prison network	370,511	8
Single hospital	467–55,492	23, 47
Multiple hospitals	1,074–25,241	53, 105
Multiple primary care practices	7,925–345,143	44, 74
Health care system	2,537–919,873	25, 48
Consortium	8,709–233,844	28, 83
Centralized anonymized repository	923–5,244,402	39, 101

Types of Data Available:

1. Demographic Data
 - Age, gender, lifestyle habits
2. Clinical Data
 - Blood pressure, cholesterol, sugar level
3. Historical Data
 - Previous diseases, surgeries

4. Diagnostic Data

- ECG results, imaging reports

Data Challenges

- Missing values
- Noisy data
- Data imbalance
- Privacy restrictions

From a business perspective, data quality directly impacts prediction reliability.

8. Business Constraints

1. Budget Limitations
2. Limited IT Infrastructure
3. Resistance from doctors
4. Regulatory approvals
5. Data availability issues

Healthcare systems must operate under strict compliance rules. Any AI system must meet regulatory standards before deployment.

9. Risk Assessment

Risk	Impact	Mitigation
Low model accuracy	Wrong diagnosis	Continuous retraining
Data breach	Legal penalty	Strong encryption
Bias in prediction	Ethical issues	Fairness testing
Doctor resistance	Low adoption	Training programs
Technical failure	Operational disruption	Backup systems

Business risk analysis ensures sustainability and reliability.

10. Cost–Benefit Analysis

Costs

- Software development
- Cloud infrastructure
- Data storage
- Staff training
- Maintenance

Benefits

- Reduced emergency costs
- Increased patient trust
- Competitive advantage
- Better insurance partnerships
- Improved operational efficiency

Return on Investment (ROI) is expected within 2–3 years if implemented effectively.

11. Success Criteria

Business Success Criteria

- 20% reduction in emergency admissions
- 15% reduction in treatment costs
- Increased patient satisfaction ratings

Technical Success Criteria

- Model accuracy > 85%
- Precision & Recall balanced
- System uptime > 99%

Operational Success Criteria

- Easy doctor adoption

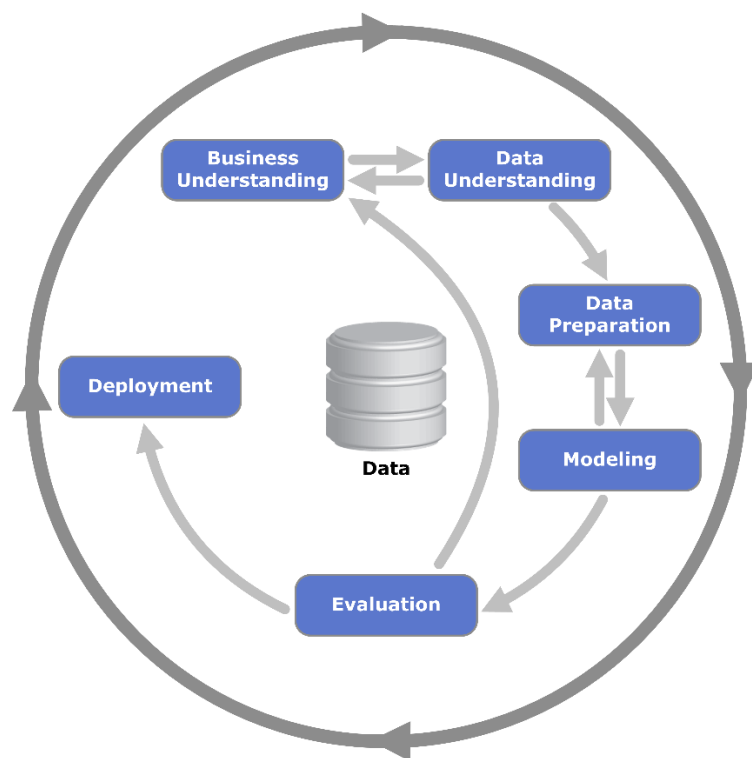
- Seamless integration
 - Real-time reporting
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12. Key Performance Indicators (KPIs)

- Prediction accuracy
- False positive rate
- Patient recovery rate
- Hospital readmission rate
- System usage rate by doctors

KPIs help measure performance after deployment.

13. Implementation Roadmap



Phase 1: Business Understanding

Define goals and requirements.

Phase 2: Data Collection

Gather patient records.

Phase 3: Data Preparation

Clean and preprocess data.

Phase 4: Model Development

Train ML algorithms.

Phase 5: Evaluation

Test performance.

Phase 6: Deployment

Integrate into hospital system.

Phase 7: Monitoring

Continuous improvement.

14. Ethical Considerations

- Patient consent for data usage
- Avoid algorithmic bias
- Transparency in predictions
- Explainable AI models
- Regular auditing

Ethics in healthcare AI is not optional; it is mandatory.

15. Conclusion

Business understanding in the healthcare domain is essential before implementing a machine learning solution for disease prediction. The goal is not just technical accuracy but delivering measurable business value, reducing healthcare costs, improving patient outcomes, and ensuring regulatory compliance.

By carefully analyzing stakeholders, objectives, constraints, risks, and success criteria, organizations can successfully implement AI-driven disease prediction systems that are sustainable, ethical, and profitable.